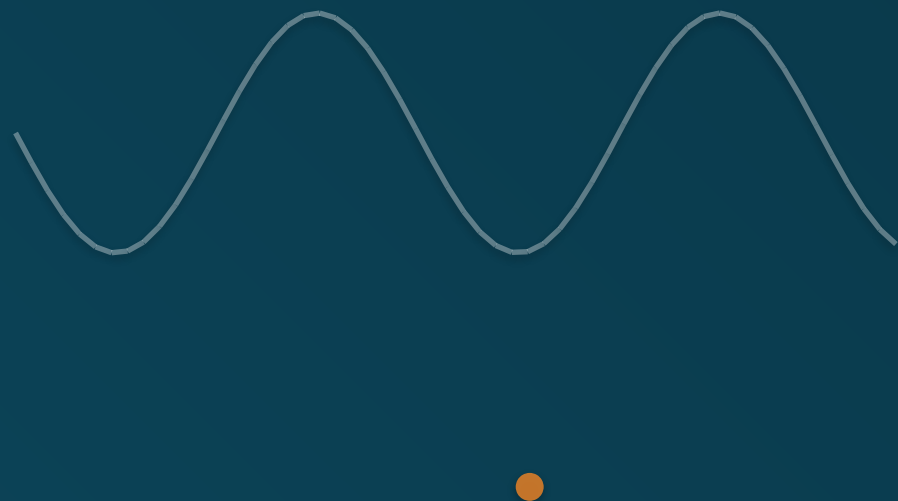


TOPIC 21

Alternating Currents

交流电



CIE 9702 · A Level Physics A2 · Paper 4

剑桥国际物理 A2 分章节讲义 · 中英双语

Learning Outcomes

学习目标

LEARNING OUTCOMES

学习目标

By the end of this topic, you will
be able to...

21.1 Characteristics of alternating currents 交流电的特征

- Period, frequency, peak value and peak-to-peak value 周期 / 频率 / 峰值
- Use $x = x_0 \sin \omega t$ for alternating current and voltage 正弦表示
- Use a CRO to measure period, frequency and peak voltage 示波器测量

21.2 Rectification and smoothing 整流与滤波

- Mean power in a resistive load; r.m.s. values 平均功率与方均根值
- Half-wave and full-wave (bridge) rectification 半波与桥式整流
- Smoothing the rectified output using a capacitor 电容滤波

Characteristics of AC

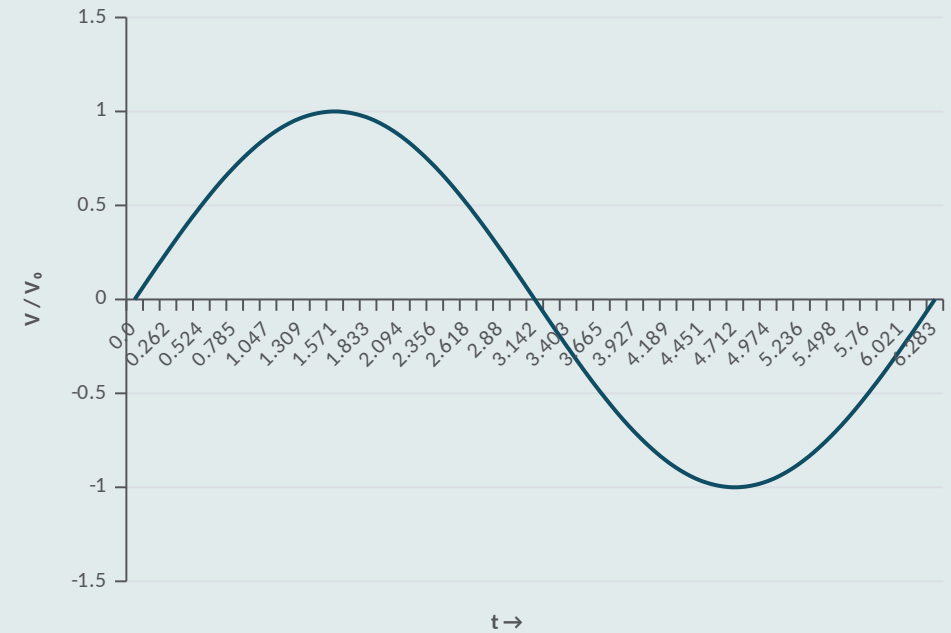
交流电的特征

An **alternating current (a.c.)** periodically reverses its direction. For a sinusoidal supply, current and voltage vary with time as a sine function.

方向周期性变化的电流；正弦交流电随时间按正弦规律变化。

$$x = x_0 \sin \omega t, \omega = 2\pi f = \frac{2\pi}{T} \quad \text{峰值 } x_0 \cdot \text{角频率 } \omega$$

- **Peak value** 峰值 : amplitude I_0 or V_0 ; **peak-to-peak** 峰 - 峰值 = $2V_0$
- Mean value over one cycle = 0 , 但传输的功率不为零 平均值零 ≠ 功率零



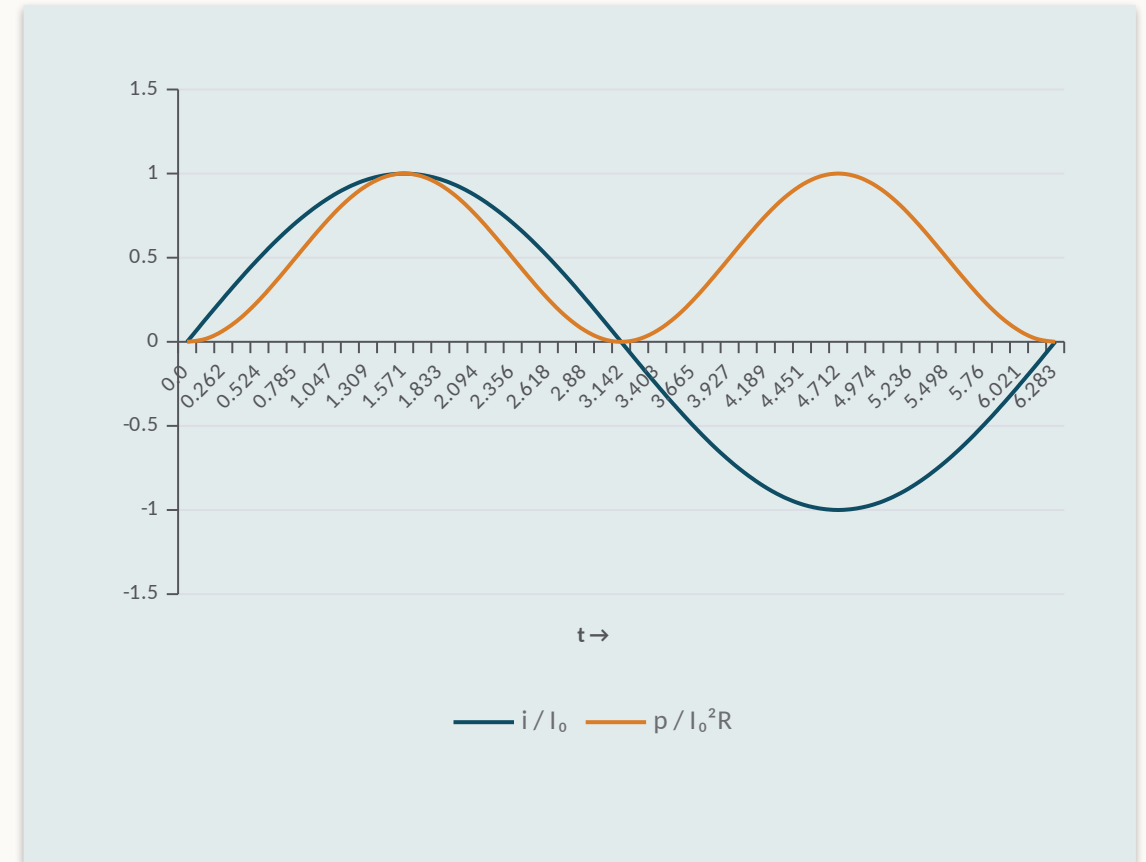
Sinusoidal a.c. voltage 正弦交流电压波形

Power in an AC Circuit

交流电路的功率

- Instantaneous power 瞬时功率: — always positive 始终为正
- Mean of $\sin^2 \omega t$ over one cycle = $\frac{1}{2} \sin^2$ 平均值为 $\frac{1}{2}$
- So the **mean power is half the maximum power** 平均功率 = 最大功率一半

$$\langle P \rangle = \frac{1}{2} P_0 = \frac{1}{2} I_0^2 R = \frac{1}{2} V_0 I_0 \quad \text{正弦交流电专用}$$



p is always positive; mean = $\frac{1}{2}$ of peak 功率恒正，均值减半

Root-Mean-Square Values

方均根值 (r.m.s.)

The **r.m.s. value** of an alternating current/voltage is the value of the **steady direct current/voltage** that would produce thermal energy at the same rate in a resistor.

方均根值：在电阻中产生相同热效应的等效直流值。

$$I_{rms} = \frac{I_0}{\sqrt{2}}, V_{rms} = \frac{V_0}{\sqrt{2}} \quad \text{仅适用于正弦波}$$

- Quoted supply voltages are r.m.s. values: 230 V mains has $V_0 = 325$ V 市电标称值是 rms
- Power formulas use r.m.s. values:

square the values 先平方



take the mean 取平均



square root 再开方

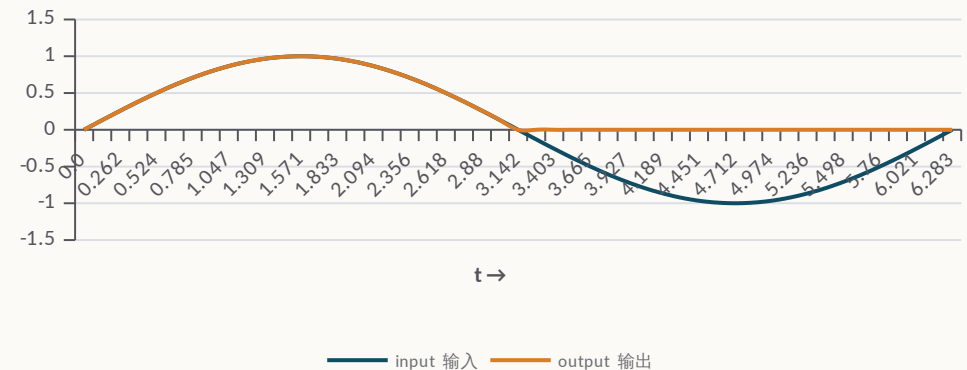
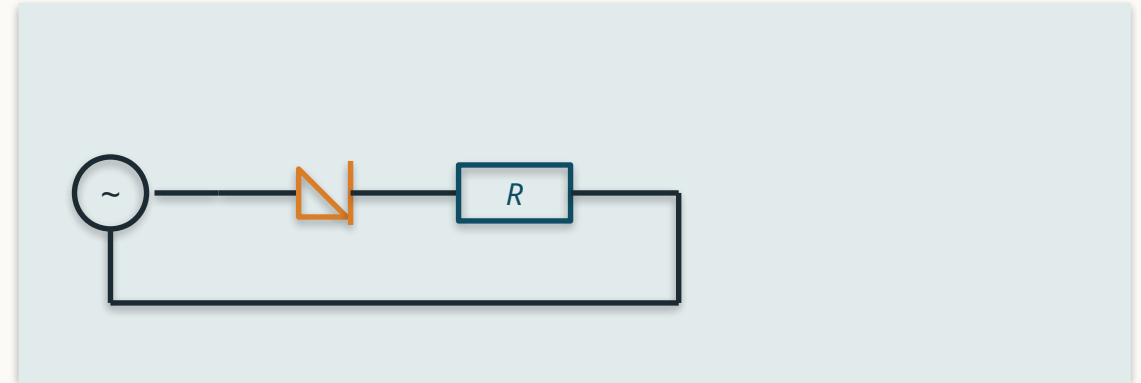
Half-Wave Rectification

半波整流

Rectification is the conversion of a.c. into d.c. A single **diode** conducts only when forward-biased, so current flows through the load during only one half of each cycle.

整流 = 交流变直流；单个二极管正向导通、反向截止。

- Output: pulsating d.c. — one half-cycle missing 输出为脉动直流
- Diode symbol: current allowed in direction of arrow 二极管符号：箭头方向允许电流
- Simple, but wastes half of every cycle 效率低，纹波大



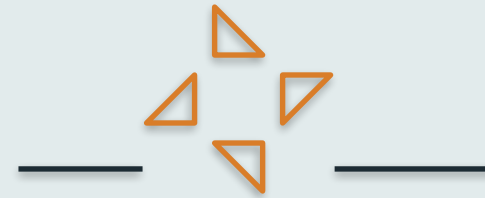
Full-Wave Rectification

桥式全波整流

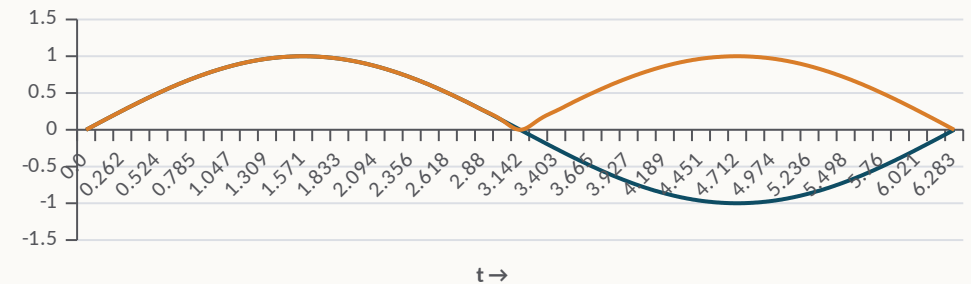
A **bridge rectifier** uses **four diodes**. Two diodes conduct on the positive half-cycle and the other two on the negative half-cycle, so current through the load is always in the same direction.

桥式整流：四只二极管，正负半周各导通两只，负载电流方向不变。

- Output uses both half-cycles 两个半周都被利用
- Mean output double that of half-wave 平均输出是半波的两倍
- Still pulsating — needs smoothing 仍需滤波平滑



bridge of 4 diodes 四二极管电桥 (菱形)



— input 输入 — output 输出

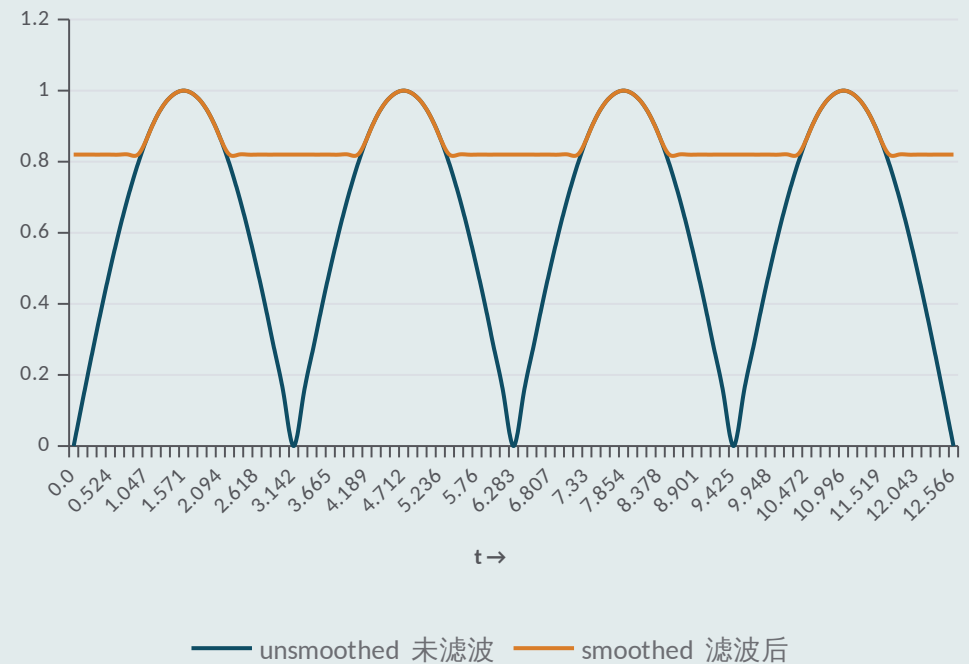
Smoothing with a Capacitor

电容滤波

A **capacitor connected in parallel with the load** charges up near the peak voltage, then discharges slowly through the load when the rectified voltage falls, keeping the output nearly steady.

与负载并联的电容：峰值时充电、下降时放电补流，使输出平缓。

- Remaining small variation = **ripple** 残余波动称纹波
- Larger C or larger load R $\Rightarrow$ slower discharge $\Rightarrow$ smaller ripple
- C、R 越大纹波越小
- Trade-off: very large C gives large surge currents through diodes 电容太大冲击电流大



Capacitor smoothing reduces ripple 滤波减小纹波

Worked Examples 21A & 21B

例题解析

WORKED EXAMPLE 21A

The variation with time t (in s) of the p.d. V (in V) of an alternating supply is

Determine: a) the peak p.d.; b) the frequency.

1. Peak value = amplitude of the sine function $\Rightarrow$
2. Compare with : rad s^{-1}
3. Hz

Answer $V_0 = 24 \text{ V}; f = 60 \text{ Hz}$

WORKED EXAMPLE 21B

An a.c. signal on a CRO shows 2 complete cycles occupying 6.0 cm, with the time-base set at 0.50 ms cm^{-1} . Find the frequency.

1. Length of one cycle $L = 6.0/2 = 3.0 \text{ cm}$
2. Period = 1.5 ms
3. Hz

Answer $f \approx 670 \text{ Hz}$

Worked Examples 21C & 21D

例题解析

WORKED EXAMPLE 21C

On a CRO with y-gain set at 5.0 mV cm^{-1} , the amplitude of the trace is 1.4 cm. Find the peak voltage of the signal.

1. Peak voltage = amplitude $\times$ y-gain
2. mV

Answer $V_0 = 7.0 \text{ mV}$

WORKED EXAMPLE 21D

A 1.5 kW heater is connected to the 230 V mains. Calculate the peak current in the heater and its resistance.

1. 230 V is the r.m.s. value: A
2. Peak current A
3. Ω

Answer $I_0 = 9.2 \text{ A}; R = 35 \Omega$

Class Practice

随堂练习

Q1 The peak value of the mains voltage in India is 325 V. Calculate the r.m.s. voltage, and the mean power dissipated in a $60\ \Omega$ resistor.

Q2 A CRO trace shows one complete cycle occupying 4.0 cm with the time-base at $2.0\ \text{ms cm}^{-1}$. Determine the frequency of the signal.

Q3 In a bridge rectifier, state which pair of diodes conducts during the positive half-cycle of the input, and explain why the output current through the load never reverses.

Answers Q1: $V_{\text{rms}} = 325/\sqrt{2} = 230\ \text{V}$, $P = 230^2/60 \approx 880\ \text{W}$ Q2: $T = 8.0\ \text{ms}$, $f = 125\ \text{Hz}$ Q3: 对角一对二极管导通；两半周电流经负载方向相同

Summary

本章小结

KEY FORMULAS 核心公式

$$x = x_0 \sin \omega t$$

正弦交流表示

$$\omega = 2\pi f$$

角频率

$$\langle P \rangle = \frac{1}{2} I_0^2 R$$

平均功率 = 峰值一半

$$I_{rms} = I_0 / \sqrt{2}$$

方均根电流

$$V_{rms} = V_0 / \sqrt{2}$$

方均根电压

$$P = V_{rms} I_{rms}$$

功率用 rms 值

KEY IDEAS 核心思想

- r.m.s. value = equivalent steady d.c. value giving the same heating effect 等效热效应直流值
- Half-wave: one diode, half of each cycle wasted 半波：一只二极管
- Bridge full-wave: four diodes, both half-cycles used 桥式：四只二极管
- Parallel capacitor smooths the output; ripple ↓ as C or R ↑ 并联电容滤波

Exam Tips

备考提示

- $I_0/\sqrt{2}$ applies to **sinusoidal** waveforms only. For square or triangular waves the r.m.s. factor is different. $1/\sqrt{2}$ 仅适用正弦波。
- Always use r.m.s. values in power calculations; using peak values overestimates power by a factor of 2. 功率计算用 rms 值。
- When sketching rectified waveforms, keep the **same period and amplitude** as the input; for half-wave show gaps, for full-wave flip the negative half up. 画波形注意周期幅值不变。
- CRO readings: period = length $\times$ time-base; peak voltage = amplitude $\times$ y-gain. Measure several cycles and average. 示波器：长度 $\times$ 档位。